# 5COSC020W DATABASE SYSTEMS – COURSEWORK SPECIFICATION (2026/27)

*Subject to external examiner approval · Version 1.2 (10 October 2026)*

## Coursework details

| | |
|---|---|
| **School** | School of Computer Science & Engineering |
| **Module** | 5COSC020W Database Systems (2026/27) |
| **Module leader** | Dr Francois Roubert |
| **Unit** | Database Systems Coursework – INDIVIDUAL COURSEWORK |
| **Weighting** | 40% of the module assessment |
| **Qualifying mark** | 30% (overall module average of at least 40%, with both components at least 30%) |
| **Description** | Evaluate an AI-generated conceptual model and produce a conceptual EERD (Part A); map a conceptual EERD to a logical ERD and evaluate relational vs NoSQL implementation approaches (Part B); document and reflect on your use of AI across the whole coursework (AI Logging and Reflections). |
| **Learning outcomes covered** | **LO1** Construct a conceptual data model that accurately captures the data requirements of a given organisation.<br>**LO2** Develop a logical database schema that effectively represents the intended implementation of the organisational data model.<br>**LO5** Critically evaluate relational and NoSQL database systems in terms of their use cases, capabilities, performance characteristics, security considerations, and environmental implications. |
| **Handed out** | Friday 02 October 2026 |
| **Submission link** | On Blackboard, under the section "NOV 2026 COURSEWORK (CWK)" |

### Deadlines

- **Thursday 15 October 2026 at 17:00:00 – Business Questions forum closes (formative; hard deadline for posting)**

  Deadline for posting Business Questions on the **Coursework Part A Business Questions forum on Blackboard:** **here** (posted on the forum, not submitted).
- **Thursday 22 October 2026 at 13:00:00 – Intermediate submission (formative, recommended; not marked)**

  **Part A, Question 1 and Question 2.1, with a draft AI log for Question 1 and Question 2.1.**
- **Thursday 5 November 2026 at 13:00:00 – Final submission (summative)**

  **FINAL SUBMISSION: Parts A and B, and AI Logging and Reflections.**

### Deliverables

**Intermediate submission (recommended) – Thursday 22 October 2026 at 13:00:00.** One report in PDF format, font size 11:

- 1 cover page, with your student and tutorial group details;
- Question 1 – Draft evaluation of the AI-generated EERD;
- Question 2.1 – Conceptual EERD (1 side, essential);
- Question 5 – Draft AI log for Question 1 and Question 2.1.

**Final submission – Thursday 5 November 2026 at 13:00:00.** One report in PDF format, font size 11:

- 1 cover page, with your student and tutorial group details;
- Question 1 – Evaluation of the AI-generated EERD (recommended maximum 1 side);
- Question 2.1 – Conceptual EERD (1 side);
- Question 2.2 – Simplified data dictionary (2 tables);
- Question 3.1 – Logical ERD (1 side);
- Question 3.2 – Mapping log (recommended maximum 1 side);
- Question 4 – Tourmato database implementation approach (recommended maximum 2 sides);
- Question 5 – AI logging and reflections (recommended maximum 2 sides).

### Feedback

- **Part A and AI log:** a global feedback document, based on a sample of intermediate submissions, will be made available to all students **on Blackboard** during Careers Week, **by Thursday 29 October 2026**. It will include guidance on what makes a good AI log.
- **Parts A and B:** individual feedback and marks via Blackboard rubrics, 25–30 working days (5–6 weeks) after the final deadline. All marks are provisional until formally agreed by the Assessment Board and the External Examiner.

### BCS accreditation criteria

- 2.1.1 Knowledge and understanding of facts, concepts, principles and theories;
- 2.1.2 Use of such knowledge in modelling and design;
- 2.2.1 Specify, design or construct computer-based systems;
- 2.3.2 Development of general transferable skills;
- 3.1.3 Knowledge of systems architecture;
- 3.2.1 Specify, deploy, verify and maintain information systems.

## Assessment regulations

For detailed information on University Assessment Regulations, including how you are assessed, penalties, late submissions and what constitutes plagiarism, please refer to: https://www.westminster.ac.uk/current-students/guides-and-policies/academic-matters/academic-regulations

**Penalty for late submission**

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark as a penalty for late submission, except for work which obtains a mark in the range 40–49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline, you will be given a mark of zero for the work in question, unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must submit a mitigating circumstances claim, giving the reason for your late or non-submission, with relevant documentary evidence. This information will be reported to the relevant Assessment Board, which will decide whether the mark of zero shall stand. Further information: https://www.westminster.ac.uk/current-students/guides-and-policies/assessment-guidelines/mitigating-circumstances-claims

**Academic Integrity**

Academic integrity means being honest, fair and responsible in your academic work. This includes completing your own assignments, exams and research, and clearly acknowledging any ideas, words or information you have used from other people or sources, for example through referencing.

If there is uncertainty regarding the authenticity of your work, a viva meeting may be held to verify the authenticity.

The meeting is a conversation between you and two academics.

A viva gives you the opportunity to:

- explain your ideas and the choices you made in your work;
- show that you understand the sources and information you used;
- show that the work was completed independently.

At the meeting, you may be asked to discuss or demonstrate parts of your work, explain how you researched and prepared it, and describe the sources or tools you used.

Before the meeting, you will have the opportunity to provide any documents or information you think are relevant. This could include draft versions of your work, research notes, plans, or other materials that show how you developed the work.

After the meeting one of the following decisions will be taken.

a. that the work will be referred for further investigation under Section 10 Academic Misconduct regulations

b. that there are no concerns regarding the submitted work, in which case the work will be marked as normal.

If you do not attend the meeting the work will be treated as a non-submission and a mark of zero will be recorded for the assessment component.

Further information: Academic misconduct | University of Westminster, London

## Overall marks allocation

| Section | Question | Marks |
|---|---|---|
| **Part A – Conceptual Modelling (Westlivingster)** | Q1 – Evaluation of the AI-generated EERD | 10 |
| | Q2 – Conceptual EERD for Westlivingster | 35 |
| **Part B – Logical Mapping and Implementation (Tourmato)** | Q3 – Logical mapping for Tourmato | 30 |
| | Q4 – Tourmato database implementation approach | 15 |
| **AI Logging and Reflections (whole coursework)** | Q5 – AI logging and reflections | 10 |
| **Total** | | **100** |

# PART A – CONCEPTUAL MODELLING: WESTLIVINGSTER (45 marks)

You have been hired as a Database Architect to undertake a database conceptual design project for Westlivingster. Your job in Part A is to evaluate an AI-generated conceptual model and to produce a high-quality **conceptual Enhanced Entity Relationship Diagram (EERD)** that supports the data needs of Westlivingster (within the relational database paradigm), documented with a simplified data dictionary.

## Part A Data Pack

### Westlivingster Data Pack Document A.1 – The Project Brief

*Authors: Board of Directors of Westlivingster Ltd*

Westlivingster is a large, purpose-built residential community made up of several apartment buildings. It was developed, and is now owned and managed, by Westlivingster Ltd, a global real-estate investor and one of the leaders in rental housing. Renting apartments is at the heart of our business, although a small number of apartments have been sold to people who now live in them.

The Westlivingster community is spread across several buildings, each with a name and an address, and each containing a number of apartments. Our apartments vary in size and number of bedrooms, and each is let at a monthly rent. Each building also offers common facilities, such as residents' lounges and other shared spaces. Every building, apartment, facility, parking lot and parking space has its own reference number, unique across the whole of Westlivingster.

Anyone who wishes to live at Westlivingster must first register with us, and from that moment we consider them one of our residents, even before they move in. When registering, each person tells us whether they wish to rent or to buy an apartment. We record their personal and contact details.

Residents who rent are our tenants. To check that tenants can afford their rent, we record their monthly salary and their credit score. Tenants rent an apartment by signing a rental agreement, which one or several tenants can sign together. Every agreement has a lead tenant: when several tenants sign together, one of them is designated as the lead tenant. The lead tenant is responsible for the agreement and is our main point of contact for that tenancy, so we record how each lead tenant prefers to be contacted. For every agreement we record its start and end dates, the deposit paid and the total cost of the tenancy. We keep all past agreements, for instance when a tenancy is renewed. For each rental agreement, we also set up the utilities contracts (such as electricity, gas and water) with the relevant suppliers. It is essential to record the start and end dates of each contract and its estimated cost.

Residents who buy an apartment are our owners. They live in the apartment they own, and only the person who purchased the apartment is registered as its owner. It is helpful for us to record the date of purchase. Owners do not sign rental agreements and arrange their own utilities.

Most buildings have their own parking lots. Some lots are reserved for our residents' apartments and others for visitors, and for each lot we record its name and capacity. Each lot is divided into a number of parking spaces. Residents may request a parking space with their apartment, but not more than one. Some of our residents' spaces are equipped with a charging point, for which we record the type and speed of the charger; the others are standard spaces. Spaces with a charging point are reserved for electric vehicles, and residents with an electric vehicle must request one. Residents who have been given a space tell us which of their vehicles will be using it.

Both our residents and our visitors can have vehicles, for which we need to record essential details. Vehicles may be electric and/or run on petrol or diesel, and we record the type of electric vehicle or the fuel used accordingly.

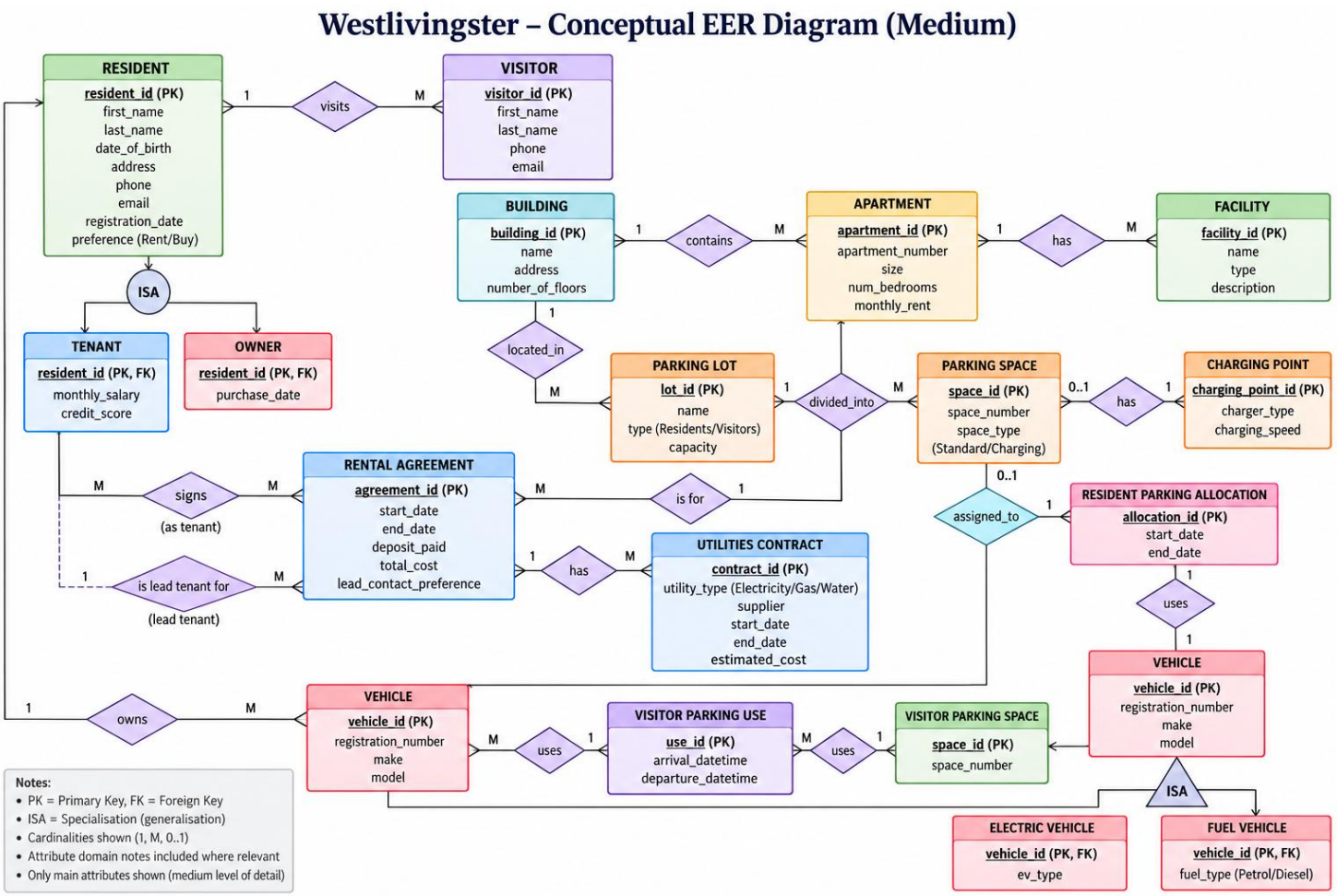
Our residents regularly receive visitors, and we keep a record of every visit: who visited which apartment, and when the visit started and ended. We record each visitor's personal and contact details. Visitors often come back, sometimes to see different residents.

Visitors coming to Westlivingster with a vehicle need to register it for one of our visitor spaces. We record every use of a visitor space, with the date and time the vehicle arrived and left.

## Westlivingster Data Pack Document A.2 – AI-Generated Conceptual EERD

*Author: ChatGPT (GPT-5.6 Sol, Medium mode), 27 September 2026.*
*Source: the Westlivingster Project Brief presented in Document A.1.*

**Prompt 1:** "Please produce a high-quality CONCEPTUAL ENHANCED ENTITY RELATIONSHIP DIAGRAM (EERD) to support the data needs of the Westlivingster business." *(followed by the brief)*
**Result 1:** generated text describing different components of the EERD.
**Prompt 2:** "Can you please create the actual diagram?"
**Result 2:** see the diagram below.



### Westlivingster Data Pack Resource A.3 – Business Questions Forum on Blackboard

Blackboard location: **Coursework Part A Business Questions**

**Deadline to post Business Questions: Thursday 15 October 2026 at 17:00:00.**

You can use this online forum to ask key Business Questions (BQs) to the Board of Directors of Westlivingster, to clarify your understanding of the business, its data needs and the components of the conceptual EERD you are building.

- BQs must be **posted on this Blackboard forum only**. They are **not submitted** through the coursework submission link, and BQs sent by email will be ignored.
- BQs must be formulated so that they are understandable to the Board of Directors, who have **no technical knowledge** of database systems. Questions directed at lecturers will not be accepted.
- BQs should be **unique**, formulated by you and only you, and **not AI-generated**.
- BQs asked after the deadline will not be considered.

An **Executive Summary** of the answers to the most relevant BQs will be made available on Blackboard after the deadline. You are strongly encouraged to use it to review and improve your conceptual EERD.

## Part A Questions

### QUESTION 1: EVALUATION OF THE AI-GENERATED EERD (10 marks)

**Evaluate** the AI-generated EERD given in Data Pack Document A.2 against the Westlivingster Project Brief (Document A.1) and the conceptual modelling techniques taught in this module. Identify **five weaknesses** of the AI-generated EERD and suggest a potential improvement for each:

- **Weaknesses 1 and 2: conceptual modelling conventions.** Issues related to the UML notation used in this module, and elements that do not belong in a conceptual model.
- **Weaknesses 3, 4 and 5: semantics.** How the diagram fails to capture the meaning of Westlivingster's data correctly: its entities, specialisations, relationships and multiplicities, or attributes. Each must refer to the brief.

### *Deliverables for Question 1*

**Weaknesses table.** Complete the table below, one row per weakness.

| # | Type | Weakness: what the AI-generated EERD shows | Evidence: the modelling convention (weaknesses 1–2) or a reference to a specific component of the Westlivingster brief (weaknesses 3–5) | Potential improvement: how the component could be better modelled |
|---|---|---|---|---|
| 1 | Conceptual modelling conventions | | | |
| 2 | Conceptual modelling conventions | | | |
| 3 | Semantics | | | |
| 4 | Semantics | | | |
| 5 | Semantics | | | |

### *Requirements for Question 1*

- Each weakness must be **specific**: name the component of the diagram concerned.
- Weaknesses 3–5 must **refer to a specific component of the Westlivingster brief** that the diagram fails to capture.
- Recommended maximum length: **one side of A4** (font size 11).

## QUESTION 2: CONCEPTUAL EERD FOR WESTLIVINGSTER (35 marks)

Produce a complete, high-quality **conceptual Enhanced Entity Relationship Diagram (EERD)** that captures the data requirements of Westlivingster, as described in the Project Brief (Document A.1) and clarified by the answers to the Business Questions (Resource A.3). Your conceptual EERD must include:

- all the relevant **entities**, with their **attributes** and **primary keys**;
- all the relevant **specialisations**, with their **participation and disjoint constraints**;
- all the relevant **relationships**, with their **multiplicities** (participation and cardinality) on both ends;
- any **attributes on relationships** the business may require.

Then document the main components of your conceptual EERD in a simplified data dictionary.

### *Deliverables for Question 2*

**2.1 Conceptual EERD.** Your complete conceptual EERD, on one page.

**2.2 Simplified data dictionary (assessed under *Clarity, notation and documentation*).** Complete the two tables below. Use the templates as given, adding as many rows as you need.

*Table 1 – Entities and specialisations*

| Entity | Specialised sub-types and constraint (if any) |
|---|---|
| *e.g. Entity_A* | *e.g. Sub_Type_1, Sub_Type_2 {constraint}* |
| | |

*Table 2 – Relationships and multiplicities*

| Entity | Multiplicity | Relationship | Multiplicity | Entity |
|---|---|---|---|---|
| *e.g. Entity_A* | *e.g. 1..1* | *e.g. relates to* | *e.g. 0..** | *e.g. Entity_B* |
| | | | | |

### *Requirements for Question 2*

- Use **UML notation only**, as taught in this module. Other notations (e.g. Chen, crow's foot) will be penalised.
- This is a **conceptual** model: do **not** include foreign keys.
- The diagram must fit on **one page** of your report and be easy to read.
- You may present the diagram on a **landscape page.**
- Prefix the names of all entities and attributes with your student ID number, starting with w (see the Key Requirements section further down).
- The simplified data dictionary tables (2.2) must be **consistent with your diagram** (2.1): every entity, sub-type, constraint, relationship and multiplicity in the tables must appear in the diagram in the same form.
- **The diagram carries the marks.** The data dictionary is assessed only for its completeness and consistency with the diagram, as part of *Clarity, notation and documentation*. The absence of a diagram will lead to zero marks for this question.

# PART B – LOGICAL MAPPING AND IMPLEMENTATION: TOURMATO (45 marks)

You have been hired as a Database Architect to undertake a database logical design project for Tourmato. Your job in Part B is to map a given conceptual EERD onto a high-quality **logical ERD** that could be implemented with a relational database management system, and to reflect on the most suitable database implementation approach for Tourmato.

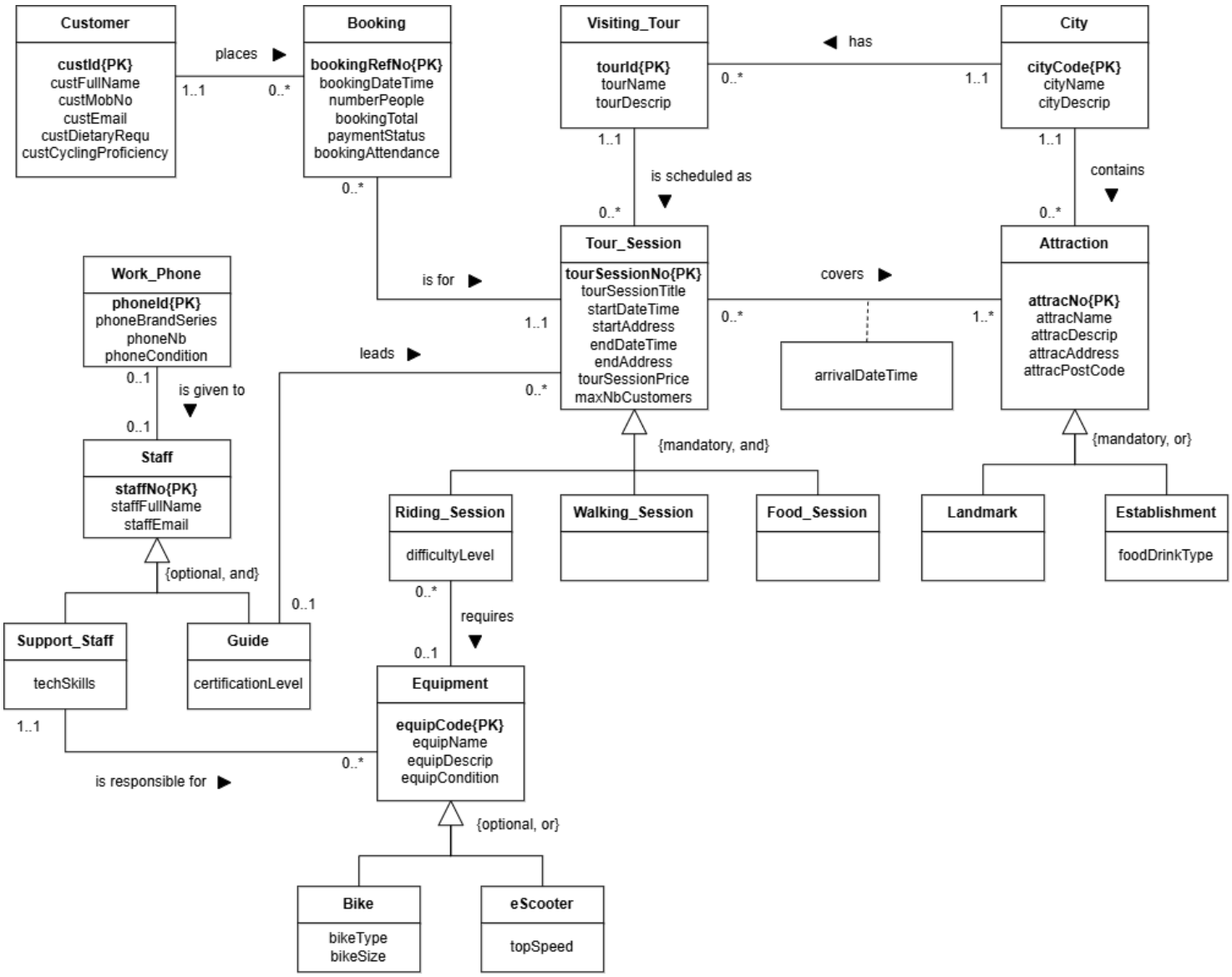
## Part B Data Pack

### Tourmato Data Pack Document B.1 – Simple Project Brief

*Authors: Board of Directors of Tourmato*

Tourmato is a tourism company offering exciting "off-the-beaten-track" visiting tours of several cities across Europe. Essentially, Tourmato takes groups of customers around European cities to visit them and experience their unique atmosphere, usually under the direction of an experienced local tour guide. To provide a range of experiences, every city covered by Tourmato offers many visiting tours, which take in its landmarks and its food and drink establishments. A tour session is a visiting tour that has been assigned a specific start date and time, start address, end date and time, end address and price. Sessions can be on foot, by bike or e-scooter, or centred on food, or a mix of these. Tourmato customers can place bookings for a variety of exciting tour sessions, depending on what they like.

### Tourmato Data Pack Document B.2 – Conceptual EERD




## Part B Questions

### QUESTION 3: LOGICAL MAPPING FOR TOURMATO (30 marks)

Map the conceptual EERD given in Data Pack Document B.2 to produce a complete **logical ERD** for Tourmato, including all the correct **tables**, **relationships**, **multiplicities**, **attributes**, **primary keys** and **foreign keys**. Using the **Logical Mapping Strategy** covered in this module, record the key mapping decisions you made: which **mapping rule** you applied to each specialisation and relationship, and how you introduced the correct foreign keys.

#### *Deliverables for Question 3*

**3.1 Logical ERD (26 marks).** Your complete logical ERD, on one page.

**3.2 Mapping Log (4 marks).** Complete the table below.

| Step | Component of the conceptual EERD | Mapping rule applied | Resulting tables, primary keys and foreign keys |
|---|---|---|---|
| *e.g. 1* | *e.g. Specialisation of Entity_A {constraint}* | *e.g. Rule n* | *e.g. tables created, PK/FK introduced* |
| | | | |

### *Requirements for Question 3*

- Use **UML notation only**, as taught in this module. Other notations will be penalised.
- The logical ERD must fit on **one page** of your report and be easy to read.
- You may present the diagram on a **landscape page.**
- Prefix the names of all tables and columns with your student ID number, starting with w (see the Key Requirements section further down).
- The mapping log (3.2) must be **consistent with your logical ERD** (3.1).
- Recommended maximum length of the mapping log: **one side of A4** (font size 11).
- **The logical ERD is what is assessed in 3.1.** The absence of a diagram will lead to zero marks for 3.1.

## QUESTION 4: TOURMATO DATABASE IMPLEMENTATION APPROACH (15 marks)

You now need to reflect on the potential implementation of a database for Tourmato, and particularly on the choice of database. Should the Tourmato database be implemented with a **relational database system**, a **NoSQL document-oriented database**, or a **combination of the two**?

To do this, you need to consider two Tourmato **workloads given to you**: typical tasks the database must perform, together with related aspects such as how often data is read or written, and how consistent it must be.

For each workload, discuss the **pros and cons** of using a relational database and of using a NoSQL document-oriented database, focusing on how well each suits the **use case** of the workload. Other aspects you may consider include **capabilities, performance characteristics, security considerations and environmental implications**; this list is not exhaustive. Support each discussion with **brief research** and with **evidence from the Tourmato conceptual EERD** given in Data Pack Document B.2.

When looking for evidence in the EERD, consider:

- how many entities and relationships must be accessed together to perform the workload;
- whether data must always be updated accurately and consistently;
- whether some entities have attributes that differ from one type to another;
- whether any of the data is sensitive.

### *Deliverables for Question 4*

**4.1. Workload table.** Complete the table below for the two given workloads.

| Workload | Relational database: pros and cons | Brief supporting research and evidence from the Tourmato conceptual EERD | NoSQL document-oriented database: pros and cons | Brief supporting research and evidence from the Tourmato conceptual EERD |
|---|---|---|---|---|
| **1. Booking a tour session:** a customer books places on a tour session and pays for them. | | | | |
| **2. Browsing tour sessions:** website visitors browse the tours offered in a city and view the details of tour sessions. | | | | |

**4.2. Concluding recommendation.** In around 100–150 words, and based on your analysis of the two workloads, state whether Tourmato should use a relational database system, a NoSQL document-oriented database or a combination of the two, and why.

### *Requirements for Question 4*

- You may present the table on a **landscape page**, and use **bullet points** within the cells.
- **Cite your sources briefly** in the table (author, year), with a short **reference list** after your recommendation.
- Recommended maximum length: **two sides of A4** (font size 11), including the table, the recommendation and the reference list.

# AI LOGGING AND REFLECTIONS – WHOLE COURSEWORK (10 marks)

*This section does not belong to Part A or Part B: it logs and reflects on your use of AI across the whole coursework.*

## QUESTION 5: AI LOGGING AND REFLECTIONS (10 marks)

Document and reflect on your use of generative AI tools across the whole coursework (Questions 1 to 4). Being transparent and critical about how you use AI is an essential professional skill, and one required for your final-year project.

Record each significant use of AI **as you go**, noting what you asked it, what it produced, how you checked it, what you did with it, and whether it was helpful. If you did not use AI for a question, say so and explain how you worked instead (see the requirements below).

### *Deliverables for Question 5*

**AI log.** Complete the table below, one row per significant use of AI, adding as many rows as you need. The first row is pre-filled for the AI-generated EERD given in Data Pack Document A.2, which you evaluate in Question 1: **every student must complete it**.

| Coursework question | AI tool | What you asked | What the AI produced and how you checked it | What you used it for | Reflection on its usefulness |
|---|---|---|---|---|---|
| **Q1** | **GPT-5.6 (Data Pack A.2, provided)** | **The prompts shown in Data Pack A.2** | | | |
| | | | | | |
| | | | | | |

*Example of a completed log entry (illustrative only: your own entries must describe your own work):*

| Coursework question | AI tool | What you asked | What the AI produced and how you checked it | What you used it for | Reflection on its usefulness |
|---|---|---|---|---|---|
| *Q4* | *ChatGPT (free version)* | *Three academic sources on how document databases handle transactions.* | *Three references. I searched for each one in the University library search: one did not exist and one was a vendor blog post.* | *I used the two real sources in my workload table and dropped the invented one.* | *A useful starting point, but every reference had to be checked. Next time I will start from the library search.* |

### *Requirements for Question 5*

- Complete the **pre-filled Data Pack A.2 row** and cover **all four questions**.
- **"No AI" entries must be fully documented.** If you did not use AI for a question, your entry must still explain what you used instead (e.g. lecture slides, mapping rules, sources), how

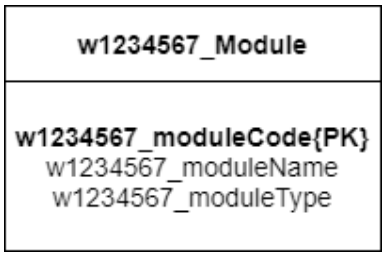
you checked your work, and where AI could have helped or been risky for that task. A bare "No AI used" will earn very few marks.

- **Marks reward the quality of your documentation and reflection, not the amount of AI you use.**
- Your log must be **consistent with your submission**. Keep your AI conversations: you may be asked to show them in a viva meeting under the University's Academic Integrity regulations. A "no AI" declaration is treated like any other log entry: if it is inconsistent with your submission, you may be invited to a viva meeting.
- **Start early.** You are strongly encouraged to include a **draft AI log for Questions 1 and 2.1** in your recommended intermediate submission (22 October). The global feedback (29 October) will include guidance on what makes a good AI log.
- Recommended maximum length: **two sides of A4** (font size 11).

## KEY REQUIREMENTS FOR THE ENTIRE COURSEWORK

- **Only UML notation is accepted** for all diagrams in this coursework, as introduced in this module.
- **Prefix all your entities, tables and attributes** with "w" followed by the 7 digits of your University ID number, both for Part A (Westlivingster conceptual EERD) and Part B (Tourmato logical ERD).
- Failure to use UML notation, or to prefix entities, tables and attributes with your ID number, will be penalised and marks will be deducted.

*Example:* if your ID number is w1234567 and you identify the entity or table "Module" with the attributes "moduleCode", "moduleName" and "moduleType", represent it as **w1234567_Module** with **w1234567_moduleCode**, **w1234567_moduleName** and **w1234567_moduleType**.



# MARKING CRITERIA

## Part A – Marking criteria

### Part A marks allocation

| Question | Marking item | Marks |
|---|---|---|
| **Q1 – Evaluation of the AI-generated EERD** | | 10 |
| **Q2 – Conceptual EERD for Westlivingster (35 marks)** | 2.1 – Entities on the conceptual EERD | 5 |
| | 2.1 – Specialisations and their constraints | 12 |
| | 2.1 – Relationships and multiplicities | 10 |
| | 2.1 – Attributes and primary keys | 4 |
| | 2.1 & 2.2 – Clarity, notation and documentation (EERD and data dictionary) | 4 |
| **Part A total** | | **45** |

### Q1 – Evaluation of the AI-generated EERD (10 marks)

| Marks | Criteria |
|---|---|
| 9–10 | Five distinct and specific weaknesses. The two conventions weaknesses are accurately justified by the UML conventions and scope of conceptual modelling taught in the module. The three semantic weaknesses each refer precisely to a specific component of the Westlivingster brief and target genuine errors in how the data is represented. Potential improvements are accurate and consistent with UML conceptual modelling. |
| 7–8 | Five valid weaknesses with mostly precise evidence. Semantic weaknesses are sound, but one may be less specific or less precisely linked to the brief. Improvements are mostly accurate. |
| 5–6 | Conventions weaknesses are sound, but semantic weaknesses are partly generic or weakly linked to the brief. Some improvements are missing or imprecise. |
| 3–4 | Mostly generic or notation-level points, with little use of the brief. Few valid improvements. |
| 0–2 | Few valid weaknesses, no evidence and no meaningful improvements. |

### Q2.1 – Entities (5 marks)

| Marks | Criteria |
|---|---|
| 5 | Most or all relevant entities are correctly identified on the conceptual EERD. |
| 4 | A good number of relevant entities are identified. |
| 3 | Some relevant entities are identified. |
| 2 | Few relevant entities are identified. |
| 0–1 | Very few or no relevant entities are identified. |

### Q2.1 – Specialisations (12 marks)

| Marks | Criteria |
|---|---|
| 11–12 | Most or all relevant specialisations are correctly identified, with their constraints (participation and disjointness) correctly captured. |
| 8–10 | A good number of relevant specialisations are identified; constraints are mostly correct. |
| 5–7 | Some relevant specialisations are identified; constraints are partly correct or missing. |
| 2–4 | Few relevant specialisations are identified; constraints are poorly captured. |
| 0–1 | Specialisations are absent or incorrect. |

### Q2.1 – Relationships and multiplicities (10 marks)

| Marks | Criteria |
|---|---|
| 9–10 | Most or all relevant relationships are correctly identified between the right entities, with multiplicities correctly capturing both participation and cardinality. |
| 7–8 | A good number of relevant relationships are identified; multiplicities are mostly correct. |
| 5–6 | Some relevant relationships are identified; multiplicities are partly correct. |
| 3–4 | Few relevant relationships are identified; multiplicities are often incorrect. |
| 0–2 | Relationships are largely absent or incorrect. |

### Q2.1 – Attributes and primary keys (4 marks)

| Marks | Criteria |
|---|---|
| 4 | All entities have meaningful attributes and valid primary keys; sub-types inherit correctly; no foreign keys appear; attributes on relationships are correctly represented. |
| 3 | Attributes and primary keys are mostly correct, with minor issues. |
| 2 | Attributes and primary keys are approximate; some key requirements are missing. |
| 0–1 | Attributes and primary keys are largely missing or incorrect. |

### Q2.1 & Q2.2 – Clarity, notation and documentation (4 marks)

| Marks | Criteria |
|---|---|
| 4 | Excellent readability; correct UML notation throughout; specialisations and constraints clearly shown; multiplicities correctly positioned; ID prefixes used throughout; a complete data dictionary (2.2), fully consistent with the diagram. |
| 3 | Good readability; correct UML notation mainly used; minor positioning or labelling issues; ID prefixes used; data dictionary mostly complete and consistent with the diagram. |
| 2 | Approximate readability and UML notation; several unclear elements; ID prefixes or the data dictionary incomplete, or several inconsistencies with the diagram. |
| 0–1 | Poor readability, or non-UML notation used; ID prefixes or the data dictionary largely missing. |

# Part B – Marking criteria

## Part B marks allocation

| Question | Marking item | Marks |
|---|---|---|
| **Q3 – Logical mapping for Tourmato (30 marks)** | 3.1 – Mapping of the specialisations | 12 |
| | 3.1 – Mapping of the many-to-many relationship | 5 |
| | 3.1 – Mapping of the one-to-many and optional one-to-one relationships | 5 |
| | 3.1 – Clarity, notation and completeness of the logical ERD | 4 |
| | 3.2 – Mapping log | 4 |
| **Q4 – Tourmato database implementation approach (15 marks)** | Workload 1: relational database (pros, cons, supporting research and evidence) | 3 |
| | Workload 1: NoSQL document-oriented database (pros, cons, supporting research and evidence) | 3 |
| | Workload 2: relational database (pros, cons, supporting research and evidence) | 3 |
| | Workload 2: NoSQL document-oriented database (pros, cons, supporting research and evidence) | 3 |
| | Concluding recommendation | 3 |
| **Part B total** | | **45** |

## Q3.1 – Mapping of the specialisations (12 marks)

| Marks | Criteria |
|---|---|
| 10–12 | All four specialisations are resolved with an excellent mapping: the right tables are created with the right attributes, flags and keys, and the relationships are kept at the correct level. |
| 7–9 | Three specialisations are resolved with a reasonable mapping. |
| 4–6 | Two specialisations are resolved with an approximate mapping. |
| 0–3 | One or no specialisation is resolved correctly. |

## Q3.1 – Mapping of the many-to-many relationship (5 marks)

| Marks | Criteria |
|---|---|
| 5 | The many-to-many relationship is fully and consistently resolved: correct link table(s), composite primary key including the foreign keys, correct relationships and multiplicities with the parent tables. |
| 4 | Correctly resolved, with one minor error. |
| 3 | Partly resolved, or several minor errors. |
| 2 | Link table(s) present, but with significant key problems. |
| 0–1 | Not resolved, or largely incorrect. |

### Q3.1 – Mapping of the one-to-many and optional one-to-one relationships (5 marks)

| Marks | Criteria |
|---|---|
| 5 | All or most relationships are correctly resolved: foreign keys are placed in the correct child tables, reference the correct parent primary keys, and the multiplicities are correct. |
| 4 | Most relationships are correctly resolved. |
| 3 | Several relationships are correctly resolved. |
| 2 | Some relationships are correctly resolved. |
| 0–1 | Few or none are resolved, or foreign keys are placed in the parent tables. |

### Q3.1 – Clarity, notation and completeness of the logical ERD (4 marks)

| Marks | Criteria |
|---|---|
| 4 | Excellent readability; correct UML notation; tables clearly laid out; primary and foreign keys clearly identified; multiplicities correctly positioned; ID prefixes used throughout; all the entities and attributes of the conceptual EERD carried into the logical ERD. |
| 3 | Good readability; correct UML notation mainly used; ID prefixes used; minor issues or a few omissions. |
| 2 | Approximate readability and notation; ID prefixes incomplete, or several tables or attributes missing. |
| 0–1 | Poor readability, or non-UML notation used; ID prefixes largely missing, or the logical ERD largely incomplete. |

### Q3.2 – Mapping log (4 marks)

| Marks | Criteria |
|---|---|
| 4 | The Logical Mapping Strategy is followed systematically; the correct mapping rule is applied and named for each specialisation and relationship type; foreign-key placement is clearly explained. |
| 3 | The strategy is followed reasonably; rules are mostly correct; minor gaps. |
| 2 | The strategy is followed approximately; some rules are missing or misapplied. |
| 0–1 | Little or no strategy; rules absent or incorrect. |

### Q4 – Each database type for each workload (3 marks each, 12 in total)

| Marks | Criteria |
|---|---|
| 3 | Relevant and specific pros **and** cons for this workload, supported by brief research with its source **and** by specific evidence from the Tourmato conceptual EERD. |
| 2 | Relevant pros and cons, but the supporting research or the EERD evidence is missing or generic; **or** well-supported but one-sided (pros only or cons only). |
| 1 | Generic or partly relevant points, with little support. |
| 0 | Missing or irrelevant. |

### Q4 – Concluding recommendation (3 marks)

| Marks | Criteria |
|---|---|
| 3 | Matches each workload to the more suitable paradigm, gives a clear overall position for Tourmato, and acknowledges a trade-off. |
| 2 | A clear, reasoned position, with one of these elements missing. |
| 1 | A position is stated with little reasoning. |
| 0 | No recommendation. |

## AI Logging and Reflections – Marking criteria

### Q5 marks allocation

| Marking item | Marks |
|---|---|
| Transparency of the AI log | 4 |
| Judgement and reflection | 6 |
| **Q5 total** | **10** |

### Q5 – Transparency (4 marks)

| Marks | Criteria |
|---|---|
| 4 | A complete log covering all four questions, with tools named and requests and outputs described clearly and specifically; the pre-filled Data Pack A.2 row completed; "no AI" entries fully documented. |
| 3 | A mostly complete log, with minor gaps or some vague entries. |
| 2 | A partial log, with several vague entries, undocumented "no AI" entries, a question missing, or the Data Pack A.2 row left incomplete. |
| 0–1 | A minimal or generic log. |

### Q5 – Judgement and reflection (6 marks)

| Marks | Criteria |
|---|---|
| 5–6 | AI outputs are consistently checked; clear decisions on what was used, modified or rejected; specific and insightful reflections on usefulness, including how it differed between tasks. |
| 3–4 | Sound checking and use, with some thin explanations; reflections partly generic or descriptive. |
| 1–2 | AI outputs mostly accepted without checking, or "no AI" entries with no account of how the work was checked; generic reflections. |
| 0 | No meaningful judgement or reflection. |

*End of document*